Reflection-even attainment for biharmonic NLS in three dimensions

A note for Diogo Oliveira e Silva

Mandel–Oliveira e Silva [1, Appendix C, Theorem 4, (4.7)] establish axial attainment and leave reflection-even attainment open. The arXiv numbering is Theorem C.2, (C.1). We prove the latter attainment for every fixed finite $p > 2$, for the original three-dimensional biharmonic symbol. For even integers $p \geq 4$, Fourier majorization of a real axial minimizer already gives equality of the axial and reflection-even levels, by [5, Lemma 5.1]; see also [4, Appendix B]. The argument below treats arbitrary finite real $p > 2$. The nonradiality conclusion concerns constrained minimizers; the earlier theorem for unrestricted ground states is [4, Theorem 1.2].

Setting and result. Write $x = (x', z) \in \mathbb{R}^2 \times \mathbb{R}$, $\Delta = \sum_{j=1}^{3} \partial_{x_j}^2$, and $H^k = H^k(\mathbb{R}^3; \mathbb{C})$. This is the Sobolev space of complex functions with square-integrable weak derivatives through order k. The notation $\|\cdot\|_r$ means the Lebesgue L^r norm. Set

$$G = O(2) \times \{1\}, \qquad G_1 = O(2) \times O(1), \qquad H^2_K = \{u \in H^2 : u \circ A = u \text{ for every } A \in K\}.$$

Here $O(j)$ is the orthogonal group on $\mathbb{R}^j$. Thus G_1 imposes axial symmetry and evenness in z. Fix $2 < p < \infty$ and $\varepsilon > 0$. Define

$$L = (\Delta + 1)^2 + \varepsilon, \qquad q(u) = \|(\Delta + 1)u\|_2^2 + \varepsilon\|u\|_2^2, \tag{1}$$

$$m = m_G = \inf_{0 \neq u \in H^2_G} \frac{q(u)}{\|u\|_p^2}, \qquad m_1 = m_{G_1} = \inf_{0 \neq u \in H^2_{G_1}} \frac{q(u)}{\|u\|_p^2}. \tag{2}$$

The physical form is $q(u) = \int_{\mathbb{R}^3} (|\Delta u|^2 - 2|\nabla u|^2 + (1 + \varepsilon)|u|^2)\, dx$. We use $\widehat{u}(\xi) = (2\pi)^{-3/2} \int e^{-i\xi \cdot x} u(x)\, dx$, extended to L^2. The Fourier symbol is $Q(\xi) = (|\xi|^2 - 1)^2 + \varepsilon$. Positivity and ellipticity give

$$c_\varepsilon (1 + |\xi|^2)^2 \leq Q(\xi) \leq C_\varepsilon (1 + |\xi|^2)^2 \quad (c_\varepsilon, C_\varepsilon > 0).$$

Thus $q^{1/2}$ is an equivalent H^2 norm, with constants depending on fixed ε. Since $H^2(\mathbb{R}^3) \subset L^p$ for every finite $p \geq 2$, both levels are positive. In [1], $m_1 = \mathrm{R}^1_\varepsilon(p)$.

Theorem 1. *For every fixed finite real $p > 2$ and every $\varepsilon > 0$,*

$$m_{G_1} < 2^{1-2/p} m_G. \tag{3}$$

Every sequence $(u_n) \subset H^2_{G_1}$ with $\|u_n\|_p = 1$ and $q(u_n) \to m_{G_1}$ is relatively compact in H^2, without translations. In particular, m_{G_1} is attained. Each normalized minimizer u satisfies $Lu = m_{G_1}|u|^{p-2}u$ weakly on all of $\mathbb{R}^3$. Thus $v = m_{G_1}^{1/(p-2)} u$ solves

$$\Delta^2 v + 2\Delta v + (1 + \varepsilon)v = |v|^{p-2} v. \tag{4}$$

For each fixed $2 < p < 4$, all these constrained minimizers are nonradial when $0 < \varepsilon < \varepsilon_0(p)$ for some $\varepsilon_0(p) > 0$. No nonradiality assertion is made for $p \geq 4$.

A real axial minimizer. The cited axial theorem applies with $d = 3$ and $s = \gamma = 2$, whose Sobolev upper exponent is infinity. Its admissibility estimates, with $r = |\xi|$, are

$$Q(r) \asymp \varepsilon + (r-1)^2 \quad (1/2 \le r \le 3/2), \qquad Q(r) \asymp 1 + r^4 \quad (r \notin [1/2, 3/2]),$$

uniformly for $0 < \varepsilon < 1$. Here $\asymp$ means upper and lower bounds by positive constant multiples. The first follows from $(r^2-1)^2 = (r-1)^2(r+1)^2$; the second follows away from $r = 1$. For any fixed $\varepsilon > 0$, including $\varepsilon \ge 1$, set $c = \max\{1, 2\varepsilon\}$. The family $g_\tau(r) = (r^2-1)^2 + c\tau$, $0 < \tau < 1$, satisfies the same admissibility estimates with τ in place of ε and constants depending on c. Applying the axial theorem at $\tau = \varepsilon/c \in (0,1)$ gives attainment for the required symbol. No uniformity in fixed ε is needed.

For $u = v + iw$ with real components, Minkowski in $L^{p/2}$ gives $\|u\|_p^2 \le \|v\|_p^2 + \|w\|_p^2$, while $q(u) = q(v) + q(w)$. A nonzero real component of an axial minimizer therefore also minimizes. Choose a real w with $\|w\|_p = 1$, $q(w) = m$. Averaging tests over compact G gives the full weak equation

$$Lw = mf, \qquad f = |w|^{p-2}w. \tag{5}$$

For completeness, when $p \ge 4$ is an even integer, put $w^\bullet = \mathcal{F}^{-1}(|\widehat{w}|)$, where $\mathcal{F}^{-1}$ is the inverse unitary Fourier transform. Reality and axial symmetry of w make $|\widehat{w}|$ invariant under full inversion and transverse inversion, hence under axial reflection. Thus $w^\bullet \in H^2_{G_1}$. Also $\widehat{w} \in L^1 \cap L^2$ by $H^2(\mathbb{R}^3)$ and Cauchy–Schwarz, so the integrability hypotheses of [5, Lemma 5.1] hold. That lemma gives $\|w^\bullet\|_p \ge \|w\|_p$, while $q(w^\bullet) = q(w)$. Hence $m \le m_1 \le q(w^\bullet)/\|w^\bullet\|_p^2 \le m$. This explains the even-integer shortcut; it does not assert that every axial minimizer is even about the origin.

The exact exponential rate. The embedding $H^2(\mathbb{R}^3) \subset C_0(\mathbb{R}^3)$ gives a bounded continuous representative vanishing at infinity. Hence $V = -m|w|^{p-2}$ is real, bounded and vanishing. Also $f \in L^2$, so $w \in H^4$. The operator $Q(-i\nabla) + V$ has eigenfunction w at energy zero, below $\operatorname{Ran} Q = [\varepsilon, \infty)$. Herbst–Skibsted [2, Theorem 1.1(1), arXiv v1] gives a positive exponential L^2 rate. The uniform interior estimate is

$$\|w\|_{H^4(B_1(x))} \le C\big(\|Lw\|_{L^2(B_2(x))} + \|w\|_{L^2(B_2(x))}\big) \le C'\|w\|_{L^2(B_2(x))},$$

where $B_r(x)$ is the radius-r ball and the constants are independent of x, since $|Lw| \le m\|w\|_\infty^{p-2}|w|$. Sobolev embedding gives $|w(x)| \le C_b e^{-b|x|}$ at every rate b for which $e^{b|x|}w \in L^2$. Thus V decays exponentially.

Use $V_1 = 0$, $V_2 = V$ in [2]; all numbered statements from that reference use arXiv v1. Theorem 1.4, with polynomial degree 4, gives a finite critical rate

$$\sigma_c = \sup\{b \ge 0 : e^{b|x|}w \in L^2\} < \infty.$$

Its hypotheses are $V_2 = O(|x|^{-2-\delta})$ and $\partial^\nu V_1 = O(|x|^{-(\delta+4+|\nu|)/2})$ for $1 \le |\nu| \le 4$, where ν is a multi-index. Both hold for every $\delta > 0$. Theorem 1.2(ii) applies because $V_2 = o(|x|^{-1/2})$ and $\partial^\nu V_1 = o(|x|^{-|\nu|})$ for every multi-index ν. No smoothness of $|w|^{p-2}$ at its zeros is needed. Equations (1.2a)–(1.2b) of [2] give real ξ and unit ω with

$$Q(\xi + i\sigma_c\omega) = 0, \qquad P_{\omega^\perp}\nabla Q(\xi + i\sigma_c\omega) = 0,$$

where $P_{\omega^\perp}$ is orthogonal projection onto $\omega^\perp$ and the polynomial extension uses $\zeta \cdot \zeta$, without conjugation. Its gradient factor $4(\zeta\cdot\zeta - 1)$ is nonzero on either characteristic quadric. Hence $\xi = t\omega$, $t^2 - \sigma_c^2 = 1$, and $2t\sigma_c = \pm\sqrt{\varepsilon}$. Therefore

$$\sigma_c = \beta = \sqrt{\frac{\sqrt{1+\varepsilon} - 1}{2}}, \qquad \alpha = \sqrt{\frac{\sqrt{1+\varepsilon} + 1}{2}}, \qquad k = \alpha + i\beta. \tag{6}$$

Indeed, $4\sigma_c^2(1 + \sigma_c^2) = \varepsilon$ has just one positive solution.

The source cannot cancel the whole characteristic surface. Put

$$b = \frac{\beta(p+2)}{2p} < \beta, \qquad \sigma_0 = \frac{\beta + (p-1)b}{2}, \qquad \beta < \sigma_0 < (p-1)b. \tag{7}$$

Then $|w(x)| \le Ce^{-b|x|}$ and $|f(x)| \le Ce^{-(p-1)b|x|}$. The ordinary Fourier–Laplace transform $F(\zeta) = \int_{\mathbb{R}^3} e^{-i\zeta\cdot x} f(x)\,dx$ is holomorphic in $|\operatorname{Im}\zeta| < (p-1)b$ and bounded on smaller closed tubes. We claim $F(k\omega)$ is not identically zero on the unit sphere S^2.

Otherwise use the holomorphic chart $(z_1, z_2) \mapsto k(z_1, z_2, -\sqrt{1 - z_1^2 - z_2^2})$ near the south pole. Here z_1, z_2 are complex coordinates, and the square root equals 1 at zero. For real z_1, z_2 near zero its image lies in kS^2. The one-variable identity theorem in each coordinate gives vanishing on an open part of $\zeta \cdot \zeta = k^2$. This is uniqueness on a maximally totally real surface. Its portion in $|\operatorname{Im}\zeta| < \sigma_0$ is connected: writing $\zeta = \xi + i\eta$, $r = |\eta|$, its points have

$$r \ge \beta, \qquad \xi = \frac{\sqrt{\varepsilon}}{2r}\frac{\eta}{r} + v, \qquad v \perp \eta, \qquad |v|^2 = 1 + r^2 - \frac{\varepsilon}{4r^2}.$$

For $\beta < r < \sigma_0$ the fibers are circles over the connected base $S^2 \times (\beta, \sigma_0)$; at $r = \beta$ they collapse continuously. Holomorphic continuation gives vanishing on this entire surface portion. Reality of f gives $F(-\overline{\zeta}) = \overline{F(\zeta)}$, hence vanishing on the conjugate quadric too.

The two quadrics are smooth and disjoint. Local holomorphic division removes both zero factors, so $T = m(2\pi)^{-3/2}F/Q$ is holomorphic on the tube $|\operatorname{Im}\zeta| < \sigma_0$ and agrees with $\widehat{w}$ on real frequencies. On smaller closed tubes $T(\xi + i\eta) = O((1 + |\xi|)^{-4})$, uniformly in η: use removability on compact sets and ellipticity at infinity. Shift Fourier inversion over cubes $[-R, R]^3$, one coordinate at a time, with bounded imaginary displacement $i\eta$ inside the tube. For fixed x, the end faces have integrals $O(R^{-2})$ as $R \to \infty$. Thus $|w(x)| \le C_\eta e^{-\eta\cdot x}$. Choose $\beta < \sigma_2 < \sigma_1 < \sigma_0$ and finitely many $|\eta| = \sigma_1$ whose directional caps cover S^2. This implies $|w(x)| \le Ce^{-\sigma_2|x|}$, contradicting (6) by taking an L^2 rate strictly between β and σ_2. The claim follows.

A positive axial interaction, including zeros at the pole. Ordinary convolution is denoted by $*$. Partial fractions of Q^{-1} and the Helmholtz kernel $e^{ik|x|}/(4\pi|x|)$ give the physical inverse kernel

$$K(x) = \frac{e^{-\beta r}\sin(\alpha r)}{4\pi\sqrt{\varepsilon}\, r}, \qquad r = |x|. \tag{8}$$

Let $S = f * f$. Exponential weighted convolution estimates give $|S(x)| \le Ce^{-\sigma_0|x|}$. For

$$U(x) = \int_{\mathbb{R}^3} \frac{e^{ik|x-y|}}{|x-y|} S(y)\,dy, \qquad C(s) = (w * Lw)(0, s),$$

absolute convergence and (5) give

$$C(s) = \frac{m^2}{4\pi\sqrt{\varepsilon}} \operatorname{Im} U(0, s), \qquad A(\omega) = \int e^{-ik\omega\cdot y} S(y)\,dy = F(k\omega)^2. \tag{9}$$

In particular, A is a nonzero axial analytic function. The square is not an absolute square.

Lemma 1. *For each fixed integer $N \ge 0$ and all sufficiently large r,*

$$U(r\omega) = \frac{e^{ikr}}{r}\left(\sum_{j=0}^{N} a_j(\omega) r^{-j} + R_N(r, \omega)\right), \quad a_0 = A, \quad a_{j+1} = \frac{(\Delta_{S^2} + j(j+1))a_j}{2ik(j+1)}. \tag{10}$$

Here Δ_{S^2} is the spherical Laplacian, and $|R_N(r, \omega)| \le C_N r^{-N-1}$ uniformly for $\omega \in S^2$, with C_N independent of r, ω.

Proof. Split the integral into $|y| \le cr$ and its complement, with fixed $0 < c < 1$. In the first part, factor out e^{ikr}/r and Taylor-expand in $h = 1/r$. The derivatives in h of $e^{ik(|r\omega - y| - r)} r/|r\omega - y|$, with its continuous extension at $h = 0$, are bounded at each fixed order by a polynomial in $|y|$ times $e^{\beta|y|}$ on $h|y| \le c$. Indeed, $|\omega - hy| \ge 1 - c$ and $r - |r\omega - y| \le |y|$. These bounds are uniform in ω and integrable against S, and give the stated Taylor remainder. Replacing truncated coefficient moments by full moments costs exponentially less. In the complement, extract $e^{-\beta r}$ using $|x - y| \ge r - |y|$; the remaining tail has rate $\sigma_0 - \beta > 0$. Near $|x - y| < 1$, use the local integrability of $|x - y|^{-1}$ and $|S(y)| \le Ce^{-\sigma_0(r-1)}$. The recurrence in (10) follows for each fixed source point before integration: conjugating $\Delta + k^2$ by e^{ikr}/r gives $v_{rr} + 2ikv_r + r^{-2}\Delta_{S^2} v$. The same exponential moments justify integrating its coefficients. No homogeneous equation outside a compact source support is assumed. □

At the south pole $\omega = -e_3$, where $e_3 = (0, 0, 1)$, write $A(\omega) = h(t)$, $t = \omega_3$. Rotational invariance makes its convergent local power series a series in $|\omega'|^2 = 1 - t^2$. Hence h is analytic at $t = -1$. Since A is nonzero and S^2 is connected, its order of vanishing there is a finite integer $J \ge 0$. Let $h_J \ne 0$ be the coefficient of $(t + 1)^J$. For $s = t + 1$,

$$\Delta_{S^2} s^j = 2j^2 s^{j-1} - j(j + 1)s^j.$$

The recurrence lowers vanishing order at most once per step. Thus $a_j(-e_3) = 0$ for $j < J$, and

$$a_J(-e_3) = \frac{J!}{(ik)^J} h_J \ne 0.$$

Now take $N = J$ in Lemma 1. After division by $e^{-\beta r} r^{-J-1}$, the error on the axis is $O(r^{-1})$. Taking $\alpha r_n + \arg a_J(-e_3) = \pi/2$ modulo 2π, with $r_n \to \infty$, gives

$$C(-r_n) \ge c_0 e^{-\beta r_n} r_n^{-J-1} > 0 \tag{11}$$

for some $c_0 > 0$ and all sufficiently large n. Kernel oscillation alone would not prove this. For example, the linear source $f = L(e^{-|x|^2})$ cancels both characteristic quadrics. The resulting potential $-L(e^{-|x|^2})/e^{-|x|^2}$ is unbounded, so the nonlinear decay argument above does not apply to that control.

Strict binding for all real $p > 2$. For real A, B, convexity gives

$$|A + B|^p \ge |A|^p + |B|^p + p(|A|^{p-2}AB + |B|^{p-2}BA) - (p + 1)|AB|^{p/2}. \tag{12}$$

Indeed, divide by the larger magnitude to reduce to $1, \pm t$, $0 \le t \le 1$. For opposite signs $(1 - t)^p \ge 1 - pt$ leaves $pt^{p-1} - t^p \ge 0$. For equal signs $(1 + t)^p \ge 1 + pt$ leaves a deficit at worst $-t^p - pt^{p-1} \ge -(p + 1)t^{p/2}$. The version without an error term fails at $p = 5/2$, $A = B = 1$, since $4\sqrt{2} - 7 < 0$.

Put $R_n = r_n/2$ and

$$a_n(x', z) = w(x', z - R_n), \quad b_n(x', z) = w(x', -z - R_n), \quad v_n = a_n + b_n.$$

Then $v_n \in H^2_{G_1}$. Let $q(a, b)$ denote the real bilinear form associated with (1). Changes of variables and transverse evenness give $I_n = q(a_n, b_n) = C(-r_n) > 0$. Equation (5) gives both nonlinear cross integrals as I_n/m and $q(v_n) = 2m + 2I_n$. For $\eta = (p - 2)/(2(p + 2))$, use $|x - R_n e_3| + |x + R_n e_3| \ge 2R_n$ and $\ge 2|x|$ to obtain

$$E_n := (p + 1) \int |a_n b_n|^{p/2} \le C_\eta e^{-pb(1-\eta)R_n}, \qquad pb(1 - \eta) = \frac{p + 6}{4}\beta > 2\beta. \tag{13}$$

Thus $E_n = o(I_n)$ by (11), for every finite J. Write $x_n = I_n/m \to 0^+$. For large n, $E_n \le (p/2)x_n$, and (12) yields

$$\|v_n\|_p^p \ge 2[1 + (3p/4)x_n], \qquad \frac{q(v_n)}{\|v_n\|_p^2} \le 2^{1-2/p} m \frac{1 + x_n}{(1 + (3p/4)x_n)^{2/p}} < 2^{1-2/p} m.$$

The last strict inequality follows because the derivative at zero of $(1 + (3p/4)x)^{2/p} - (1 + x)$ is $1/2$. This proves (3). Infinitely separated reflected copies give only the threshold $2^{1-2/p}m$, with no strict gain.

Every complex minimizing sequence. Coercivity bounds a normalized minimizing sequence in H^2, hence in L^2 and L^∞. If $\sup_y \int_{B_1(y)} |u_n|^2 \to 0$, Lions' vanishing lemma [6, Lemma I.1, p. 231] applied to $|u_n| \in H^1$ gives convergence to zero in L^3. Interpolating with L^2 for $2 < p < 3$, and with L^∞ for $p \geq 3$, contradicts normalization for every finite p. Local Rellich compactness gives a nonzero translated weak profile. Transverse centers cannot diverge: arbitrarily many rotated disjoint balls would then have the same positive local L^2 mass. This contradicts the global L^2 bound. Absorb bounded transverse centers into a larger ball, leaving only axial translations. In each frame used below, local Rellich compactness and subsequence extraction give almost-everywhere convergence for the Brezis–Lieb lemma [7, Theorem 1].

Put $\theta = 2/p \in (0,1)$. If the axial centers are bounded, the unshifted weak limit $u \in H^2_{G_1}$ has mass $a = \|u\|_p^p > 0$. Hilbert norm splitting and the complex Brezis–Lieb lemma give

$$1 = a + \|u_n - u\|_p^p + o(1), \qquad q(u_n) = q(u) + q(u_n - u) + o(1).$$

Hence $m_1 \geq m_1[a^\theta + (1-a)^\theta]$. Strict concavity forces $a = 1$, and the residual q norm tends to zero.

Otherwise extract $|t_n| \to \infty$. Let $T_t u(x', z) = u(x', z - t)$ and $\mathcal{S}u(x', z) = u(x', -z)$, with $T_{-t_n} u_n \rightharpoonup u \in H^2_G$. Reflection supplies two separated profiles $T_{t_n} u$ and $T_{-t_n}\mathcal{S}u$, each of mass $a = \|u\|_p^p > 0$. Set $v_n = u_n - T_{t_n} u$ and $r_n = v_n - T_{-t_n}\mathcal{S}u$. The residual r_n is exactly G_1 invariant, also for complex u: reflection uses no conjugation. Translates of a fixed profile tend weakly to zero as the centers separate. Two successive splittings give

$$\begin{aligned} 1 &= a + \|v_n\|_p^p + o(1), & q(u_n) &= q(u) + q(v_n) + o(1), \\ \|v_n\|_p^p &= a + \|r_n\|_p^p + o(1), & q(v_n) &= q(u) + q(r_n) + o(1). \end{aligned}$$

Thus $0 < 2a \leq 1$, and

$$m_1 \geq 2ma^\theta + m_1(1-2a)^\theta > m_1[(2a)^\theta + (1-2a)^\theta] \geq m_1,$$

by (3), a contradiction. The argument applies to every subsequence, proving relative H^2 compactness. Compact G_1 averaging gives the full Euler equation, and scaling gives (4).

Nonradiality and limits of the conclusion. Once attainment is known, nonradiality is an immediate corollary of the existing quotient estimates. Let m_{rad} be the quotient infimum over radial H^2 functions. Theorem 1.3, (1.12)–(1.13), of [1], with $d = 3$, $k = 1$, and $s = \gamma = 2$, gives for each fixed $2 < p < 4$ and small ε

$$m_1 \leq C_p \varepsilon^{2/p}, \qquad m_{\mathrm{rad}} \geq \begin{cases} c_p \varepsilon^{3/p - 1/2}, & 2 < p < 3, \\ c_3 \varepsilon^{1/2} |\log \varepsilon|^{-2/3}, & p = 3, \\ c_p \varepsilon^{1/2}, & 3 < p < 4. \end{cases} \tag{14}$$

Here $c_p, C_p > 0$ are independent of ε, but may depend on p. The exponent differences are $1/2 - 1/p > 0$ below 3 and $2/p - 1/2 > 0$ above 3; the logarithm at 3 does not offset the positive power difference. Hence $m_1/m_{\mathrm{rad}} \to 0$ at each fixed $2 < p < 4$. This proves the nonradiality assertion in Theorem 1 using the paper's existing level estimates.

All quantifiers in the attainment theorem are at fixed finite $p > 2$ and fixed $\varepsilon > 0$. There is no assertion at $p = \infty$, no common quantitative binding margin, and no common or explicit small-ε cutoff. There is no claimed uniformity as $p \downarrow 2$ or $\varepsilon \downarrow 0$. No sharp level or sharp constants are computed. Nothing about nonradiality for $p \geq 4$ follows here. No classification or uniqueness of unrestricted ground states is obtained. In particular, the theorem does not decide whether unrestricted ground states have G_1 symmetry up to coordinates, or prove the separate strict comparison $m_{\mathrm{all}} < m_{G_1}$, where m_{all} is the quotient infimum over all nonzero H^2 functions.

Klein–Roudenko [3, Remark 3.12 (Attainment)] explicitly list G_1 attainment and its strict comparison with the unrestricted level as unknown in dimension 2. This note concerns dimension 3 and does not settle

that two-dimensional problem. The proof uses $H^2(\mathbb{R}^3) \subset L^\infty$ and integrability of $(1+|\xi|)^{-4}$ in Fourier inversion; neither holds as stated in dimension $d \geq 4$. Extension would require additional regularity and dimension-dependent kernel and contour estimates, and is not proved here. The even-integer shortcut above is separate from the argument for arbitrary real p. No extension to other symbols or priority judgment is claimed.

The holomorphic division argument, the expansion for an exponentially decaying noncompact source, and compactness of arbitrary complex sequences are analytic arguments, not consequences of finite numerical tests. The scalar and linear controls above exclude shortcuts; none is asserted to be a nonlinear minimizer.

References

[1] R. Mandel and D. Oliveira e Silva, *Block-radial symmetry breaking for ground states of biharmonic NLS*, Calc. Var. Partial Differential Equations **63** (2024), article 45. doi:10.1007/s00526-023-02654-9; arXiv:2306.03720v1.

[2] I. Herbst and E. Skibsted, *Decay of eigenfunctions of elliptic PDE's, I*, Adv. Math. **270** (2015), 138–180, doi:10.1016/j.aim.2014.11.001. Numbered statements here refer to arXiv:1306.6878v1 (2013): Theorems 1.1(1), 1.2(ii), 1.4; equations (1.2a)–(1.2b).

[3] C. Klein and S. Roudenko, *Ground states to bi-harmonic nonlinear Schrödinger equations with competing dispersion*, arXiv:2609.13783v1, 12 September 2026, Remark 3.12 (Attainment).

[4] E. Lenzmann and T. Weth, *Symmetry breaking for ground states of biharmonic NLS via Fourier extension estimates*, arXiv:2110.10782v2 (2021), Theorem 1.2 and Appendix B.

[5] E. Lenzmann and J. Sok, *A sharp rearrangement principle in Fourier space and symmetry results for PDEs with arbitrary order*, arXiv:1805.06294v4 (2019), Lemma 5.1.

[6] P.-L. Lions, *The concentration-compactness principle in the calculus of variations. The locally compact case, part 2*, Ann. Inst. H. Poincaré Anal. Non Linéaire **1** (1984), 223–283, Lemma I.1, p. 231.

[7] H. Brezis and E. Lieb, *A relation between pointwise convergence of functions and convergence of functionals*, Proc. Amer. Math. Soc. **88** (1983), 486–490, Theorem 1.

Checker. Run `python3 -I check_diogo_db51.py` with Python 3.9 or later and the standard library. The script verifies the encoded rational identities, sign margins, recurrence coefficients, and algebraic controls. It does not read this note or verify its analytic lemmas or cited theorems. In particular, it does not derive the kernel normalization or prove the reflection and compactness arguments. SHA-256:

```
4a2174b31dcd1f285c7acc5fd41e6baf
5a4a0a2e90f44aa7c4a795c6e999f7c1
```